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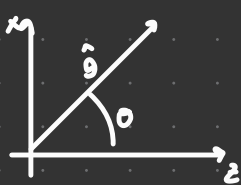
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SPIN 1/2 PARTICLE

$$S_\theta = \sin\theta S_y + \cos\theta S_x$$



$$S_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

1. WHAT ARE THE POSSIBLE OUTCOMES of a SPIN MEASUREMENT ALONG θ ?

2. WHAT ARE THE STATE ASSOCIATED WITH THESE OUTCOMES?

$$S_\theta = \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

$$|\psi_+\rangle, \lambda_+ = +1$$

$$S_\theta = \hbar/2, -\hbar/2$$

$$\alpha_+ |z_+\rangle + \alpha_- |z_-\rangle$$

TRIVIAL SOL.

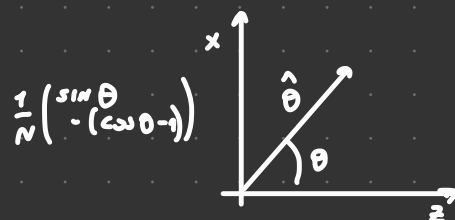
$$\begin{pmatrix} \cos\theta - \lambda & \sin\theta \\ \sin\theta & -\cos\theta - \lambda \end{pmatrix} \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} = 0$$

$$(\cos\theta - 1)\alpha_+ + \sin\theta\alpha_- = 0$$

$$\begin{cases} \alpha_+ = \sin\theta \\ \alpha_- = -(\cos\theta - 1) \end{cases}$$

$$\frac{1}{\sqrt{\alpha_+^2 + \alpha_-^2}} \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} \rightarrow \frac{1}{N} \begin{pmatrix} \sin\theta \\ -(\cos\theta - 1) \end{pmatrix}$$

$$\frac{1}{N} = \sqrt{(\cos\theta - 1)^2 + \sin^2\theta}$$



$$\theta = 0 \Rightarrow S_\theta = S_z \quad |\psi_+\rangle = |z_+\rangle \quad |\psi_-\rangle = |z_-\rangle$$

$$|\psi_+\rangle = \frac{1}{N}$$

$$\theta = 0 \quad 2\alpha_- = 0 \Rightarrow \alpha_- = 0 \quad |\psi_+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow |z_+\rangle$$

EX 3.53

WE MENTIONED THAT THE VARIANCE of an OBSERVABLE A CAN BE CALCULATED AS $\sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$

SPIN 1/2 PARTICLE



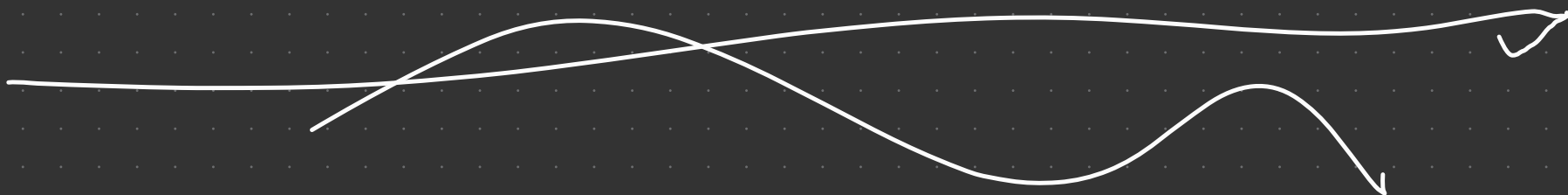
THE AVERAGE of the DISTRIBUTION IS 0
 $\langle S_z \rangle = 0$

A PECULIAR, YET IMPORTANT SITUATION, IS THAT IN WHICH TWO or MORE OUTCOMES of a MEASUREMENT of an OBSERVABLE A ARE IDENTICAL

E.g. CONSIDER A THREE-LEVEL ATOM WITH $E_1 \neq E_2 = E_3$

$$H = E_1 |1\rangle\langle 1| + E_2 (|2u\rangle\langle 2u| + |2d\rangle\langle 2d|)$$

$$H|1\rangle = E_1|1\rangle \quad H|2u\rangle = E_2|2u\rangle \quad H|2d\rangle = E_2|2d\rangle$$



COMPATIBLE OBSERVABLE TWO OBSERVABLE A AND B ARE SAID TO BE COMPATIBLE IF THERE IS A BASIS MEASUREMENT THAT CAN DETERMINE THE VALUES OF BOTH A AND B TOGETHER

THUS TWO OBSERVABLE ARE COMPATIBLE IF THEY HAVE A COMMON EIGENVALUE BASIS

$$A = \sum_N A_N |n\rangle \langle n|$$

CONSIDER THE CASE OF A SPIN $1/2$ PARTICLE AND THE FOLLOWING OPERATORS

$S_z = \frac{\hbar}{2} \hat{z}$ WHICH GIVES THE VALUE OF THE SPIN COMPONENT ALONG z

$$S = \sqrt{S_x^2 + S_y^2 + S_z^2} = \frac{\sqrt{3}}{2} \hbar \hat{1} \text{ WHICH GIVES THE SPIN MODULUS}$$

$$|\psi\rangle = \alpha |z_+\rangle + \beta |z_-\rangle$$

$$P\left(\overset{S_z}{+\hbar/2}; \overset{S}{\sqrt{3}/2}\hbar\right) \quad P(|\psi\rangle)(|z_+\rangle)$$